- Most changes are marked in yellow.
- The timing is now exactly as in Chapter 9, i.e., the economy starts at time 0 (when all agents also trade under time 0 trading arrangement).
- Enhanced clarity about our clearinghouse equilibrium concept.

Chapter 10
Overlapping Generations

This chapter describes the pure exchange overlapping generations model of Paul Samuelson (1958). We begin with an abstract presentation that treats the overlapping generations model as a special case of the chapter 9 general equilibrium model with complete markets and all trades occurring at time 0. There is a special type of heterogeneity across agents: each person cares about consumption only at two adjacent dates, and the set of individuals who care about consumption at a particular date includes some who care about consumption one period earlier and others who care about consumption one period later. We shall study how this special preference and demographic pattern affects some outcomes of the chapter 9 model.

Imagining complete markets with time 0 trading in an overlapping generations model strains credulity by imagining that equilibrium price and quantity sequences are set at time 0, before most of the people who are to execute trades have been born. For that reason, most applied work with the overlapping generations model assumes a sequential-trading arrangement, like the sequential trade in Arrow securities described in chapter 9. In a sequential-trading arrangement, all trades are agreed to only by living agents.

Despite its implausibility, we use the abstraction of complete markets with time 0 trading to better understand possible peculiarities in an overlapping generations model such as multiplicity of equilibria and Pareto inefficient equilibrium allocations, which could not happen in the chapter 9 general equilibrium model. This will also demonstrate that the existence of such peculiarities in an overlapping generations model does not hinge upon agents alive today not being able to trade with future generations since we assume that everyone can trade with anyone else at time 0.[1]

Later in this chapter, we use versions of the model with sequential trading to tell how the overlapping generations model provides a framework for thinking about equilibria with government debt, valued fiat currency, intergenerational transfers, and fiscal policy.

[1] To be explained in section 10.3, under time 0 trading, we will switch from a Walrasian equilibrium to an alternative equilibrium concept that entails a clearinghouse.

10.1. Preferences and endowments

Time is discrete, starts at $t = 0$, and lasts forever, so $t = 0, 1, 2, \ldots$. In each period, there is a large number N of identical agents born who live for two periods. The assumption of a large cohort N is needed for the analysis of a competitive equilibrium in which agents take equilibrium prices as given. For expositional simplicity we will refer to a representative agent in each generation. Thus, there is an infinity of (representative) agents, one for each generation, named $i = -1, 0, 1, \ldots$. We can also regard i as agent i's period of birth.

There is a single good at each date. The good is not storable. There is no uncertainty. Each agent has a strictly concave, twice continuously differentiable, one-period utility function $u(c)$ that is strictly increasing in consumption c of the one good. Agent i orders consumption stream $\mathbf{c}^i = \{c_t^i\}_{t=0}^{\infty}$ according to the special utility functional

$$U^i(\mathbf{c}^i) = u(c_i^i) + u(c_{i+1}^i), \quad i \geq 0, \tag{10.1.1a}$$

$$U^{-1}(\mathbf{c}^{-1}) = u(c_0^{-1}). \tag{10.1.1b}$$

As a superscript, i denotes a person's birth date; as a subscript, it denotes the current date. Notice that agent $i \geq 0$ only wants goods dated i and $i+1$, while an agent born at $i = -1$, before the model starts at date $i = 0$, wants consumption only at time 0. Equations (10.1.1) tell us that agents want to consume only when they are alive in periods $t \geq 0$.

Each consumer $i \geq -1$ has an endowment sequence $\mathbf{y}^i = \{y_t^i\}_{t=0}^{\infty}$.

Definitions: An *allocation* is a list $\{\mathbf{c}^i\}_{i\geq -1} = \{\{c_t^i\}_{t\geq 0}\}_{i\geq -1}$ of each agent i's consumption at every date t. A *feasible allocation* satisfies

$$\sum_i c_t^i \leq \sum_i y_t^i \tag{10.1.2}$$

for all $t \geq 0$.

In most of our analysis, we will assume that consumers are endowed with goods only when they are alive. That is, each consumer $i \geq 0$ has an endowment sequence $\mathbf{y}^i$ satisfying $y_i^i \geq 0$, $y_{i+1}^i \geq 0$, $y_t^i = 0 \ \forall t \neq i$ or $i+1$; whereas the inital old consumer $i = -1$ has a single endowment $y_0^{-1} \geq 0$. Evidently an allocation is then feasible if for all $t \geq 0$,

$$c_t^t + c_t^{t-1} \leq y_t^t + y_t^{t-1}. \tag{10.1.3}$$

Furthermore, we will mostly focus on the case of stationary endowments when $y_i^i = y_y$ and $y_i^{i-1} = y_o$ for all $i \geq 0$. Here thee subscript y denotes young and o denotes old.

10.2. Dynamic (in)efficiency

Pareto inefficient equilibria in this chapter arise from intertemporal inefficiencies. Therefore, such equilibria in an overlapping generations model are commonly refered to as being *dynamically inefficient*. Under the preference specification U^i in (10.1.1) and stationary endowments, $(y_i^i, y_i^{i-1}) = (y_y, y_o)$ for all $i \geq 0$ and $y_t^i = 0$ otherwise, a multiplicity of equilibria and dynamically inefficient equilibria will exist if and only if $y_o < y_y$. If $y_o \geq y_y$, there is a unique equilibrium that is also Pareto efficient, as in the chapter 9 general equilibrium model.

10.3. Time 0 trading

As in chapter 9, we begin by analyzing an economy in which all trades take place at time 0. But instead of a "Walrasian auctioneer" posting equilibrium prices we assume that there is a "clearinghouse" that posts prices, monitors demands and supplies for goods in all periods, and verifies that each person's single time 0 budget constraint is satisfied. An equilibrium price vector assures that for periods $t \geq 1$ aggregate endowments in that period are consumed by living agents, but it can turn out that in period 0 the aggregate endowment exceeds aggregate consumption which would mean that the clearinghouse ends up with some time 0 good not allocated. If that happens we simply assume that the excess supply of the time 0 good, denoted $\Omega_0 \geq 0$, is left to perish because no agent desiring those goods can afford to buy them. Hence, the market clearing conditions under the alternative equilibrium concept with a clearinghouse are

$$c_t^t + c_t^{t-1} = y_t^t + y_t^{t-1}, \quad \forall t \geq 1; \tag{10.3.1a}$$

$$c_0^0 + c_0^{-1} + \Omega_0 = y_0^0 + y_0^{-1}, \tag{10.3.1b}$$

where $\Omega_0 \geq 0$.

The peculiarity with excess supply of time 0 goods cannot happen in the chapter 9 general equilibrium model. Hence, it would not matter if we compute a Walrasian equilibrium or use our alternative equilibrium concept that entails a clearinghouse for that economy. Both equilibrium concepts lead to the same unique equilibrium allocation as we characterized in chapter 9. "Walras Law" holds in that economy. Restated in terms of our present setup, Walras Law would say that if all time 0 markets for goods dated $t \geq 1$ clear so must the remaining market $t = 0$. But as we will demonstrate, Walras' Law can be violated in an overlapping generations model. So here it matters that we adopt the alternative equilibrium concept with a clearinghouse which enables us to analyze a richer set of possible outcomes rather than being stymied by the restrictiveness of a Walrasian equilibrium. In fact, inspired by those additional outcomes that emerge under the alternative equilibrium concept with a clearinghouse, we will later study sequential trading equilibria in which fiat money and unbacked government debt might be valued in an overlapping generations model. Outcomes that could never happen in the chapter 9 general equilibrium model.

10.3.1. Agents' optimization

Thus, at time 0, there are complete markets in time $t \geq 0$ consumption goods with prices $\{q_t^0\}_{t\geq 0}$.[2] Agent i's budget constraint is

$$\sum_{t=0}^{\infty} q_t^0 c_t^i \leq \sum_{t=0}^{\infty} q_t^0 y_t^i. \tag{10.3.2}$$

Letting μ^i be a Lagrange multiplier attached to the budget constraint of consumer $i \geq 0$, the consumer's first-order conditions are

$$\mu^i q_i^0 = u'(c_i^i), \tag{10.3.3a}$$

$$\mu^i q_{i+1}^0 = u'(c_{i+1}^i), \tag{10.3.3b}$$

$$c_t^i = 0 \text{ if } t \notin \{i, i+1\}. \tag{10.3.3c}$$

[2] As shown in chapter 9, only relative prices are determined in an equilibrium. For absolute prices, we need to choose a numeraire. Here we have chosen the good at time 0 as the numeraire. Hence, using our notation in chapter 9, q_t^0 is the price of one good to be delivered at time t in terms of the good at time 0.

The consumer's Euler equation can be deduced from equations (10.3.3a) and (10.3.3b),

$$\frac{u'(c_{i+1}^i)}{u'(c_i^i)} = \frac{q_{i+1}^0}{q_i^0} \equiv \alpha_i = \frac{1}{R_i}, \tag{10.3.4}$$

where α_i is the price of one good dated $i+1$ in terms of the good dated i.[3] Later, in a sequential trading equilibrium, this will be the inverse of the one-period gross interest rate R_i between periods i and $i+1$.

Regarding the initial old agent $i = -1$, who is alive only in the first period of the economy, the optimal consumption choice is simply to consume his entire wealth in period 0. Under our specialization of endowment streams, the initial old consumer's budget constraint is $q_0^0 c_0^{-1} \leq q_0^0 y_0^{-1}$ and hence, the optimal consumption choice is $c_0^{-1} = y_0^{-1}$.

10.3.2. Example equilibria

Let $\epsilon \in (0, .5)$. Endowments are

$$\begin{aligned} y_i^i &= 1 - \epsilon, \ \forall i \geq 0, \\ y_{i+1}^i &= \epsilon, \ \forall i \geq -1, \\ y_t^i &= 0 \text{ otherwise.} \end{aligned} \tag{10.3.5}$$

This economy has many equilibria. We describe two stationary equilibria now and nonstationary equilibria later. An equilibrium is said to be a stationary equilibrium if its allocation is stationary in the following sense.

Definition: An allocation is *stationary* if $c_i^i = c_y$ and $c_{i+1}^i = c_o$ for all $i \geq 0$.

Here the subscript y denotes young and o denotes old. Note that we do not require that $c_0^{-1} = c_o$. In fact, as explained above, the equilibrium consumption of the initial old agent $i = -1$ will be $c_0^{-1} = y_0^{-1} = \epsilon$ in all equilibria, both stationary and nonstationary equilibria.

We can use a guess-and-verify method to confirm two stationary equilibria. Specifically, we proceed by first guessing the equilibrium allocation for agents $i \geq 0$, which amounts to guessing c_y and c_o in a stationary equilibrium. Given that guess, we use Euler equation (10.3.4) to deduce at which relative price

[3] If we had used our notation in chapter 9, α_t would have been written as q_{t+1}^t.

q^0_{i+1}/q^0_i such a consumption profile (c_y, c_o) would be consistent with utility maximization by agent $i \geq 0$. Next, given that deduced relative price and the guess of an allocation, we check that budget constraint (10.3.2) holds with equality for each agent $i \geq 0$, which here becomes

$$c_y + \frac{q^0_{i+1}}{q^0_i} c_o = (1-\epsilon) + \frac{q^0_{i+1}}{q^0_i}\epsilon, \tag{10.3.6}$$

where we have divided both sides of the budget constraint by q^0_i and hence, changed the numeraire to be the good dated i. Finally, we verify that markets clear in accordance with equations (10.3.1).

So by using this guess-and-verify method, we can confirm the following two stationary equilibria.

1. Equilibrium H: a high-interest-rate equilibrium. We guess a stationary equilibrium allocation $c_y = c_o = .5$ and $c_0^{-1} = \epsilon$. To be consistent with utility maximization, Euler equation (10.3.4) must be satisfied and hence, the relative price would have to be $q^0_{i+1}/q^0_i = u'(.5)/u'(.5) = 1 = \alpha_H$. Given our normalization $q^0_0 = 1$, the sequence of absolute prices becomes $q^0_t = 1$ for all $t \geq 0$. We can then verify that budget constraint (10.3.6) holds with equality for each agent $i \geq 0$, and trivially for the initial old agent $i = -1$ who simply consumes his endowment in period 0, $c_0^{-1} = \epsilon$. Concerning market clearing conditions, equation (10.3.1a) holds since aggregate consumption equals aggregate endowments in periods $t \geq 1$, while equation (10.3.1b) yields excess supply of goods dated 0,

$$\Omega_0 = y^0_0 + y_0^{-1} - c^0_0 - c_0^{-1} = .5 - \epsilon > 0,$$

   where the quantity $\Omega_0 > 0$ of goods dated 0 is left to perish since no agent desiring those goods can afford to buy them.

2. Equilibrium L: a low-interest-rate equilibrium. We guess a stationary equilibrium allocation $c_y = 1-\epsilon$, $c_o = \epsilon$, and $c_0^{-1} = \epsilon$. For that consumption profile to be consistent with utility maximization, the relative price would have to be $q^0_{i+1}/q^0_i = u'(\epsilon)/u'(1-\epsilon) = \alpha_L > 1$, i.e., Euler equation (10.3.4) must hold for all $i \geq 0$. Given our normalization $q^0_0 = 1$, the sequence of absolute prices becomes $q^0_t = (\alpha_L)^t$ for all $t \geq 0$. Since all agents are simply consuming their endowments, $c^i_t = y^i_t$ for all i, t, budget constraints (10.3.2) and market clearing conditions (10.3.1) are trivially satisfied, and

there is no excess supply of goods dated 0, $\Omega_0 = 0$. This equilibrium is autarkic, with equilibrium prices eradicating all incentives to trade.

To understand why we call them high- and low-interest-rate equilibria, recall that the relative price α_t in expression (10.3.4) would be the inverse of the one-period gross interest rate R_t in a sequential trading equilibrium. Thus, the interest rates associated with the high- and low-interest-rate equilibrium are $R_H = 1$ and $R_L = (\alpha_L)^{-1} < 1$, respectively. Soon we shall construct equilibria in which R_t varies over time t.

10.3.3. Relation to welfare theorems

Equilibrium H Pareto dominates equilibrium L. In equilibrium H, every generation after the initial old one is better off and no generation is worse off than in equilibrium L. The equilibrium H allocation is peculiar because some of the time 0 good is not consumed, leaving room to set up a giveaway program to the initial old that makes them better off. We shall soon learn how an institution of either fiat money or of perpetual government debt can accomplish this purpose.[4]

With equality between supply and demand in all periods, equilibrium L is also a Walrasian equilibrum. But, evidently, this competitive equilibrium fails to satisfy the assumptions needed to deliver the first fundamental theorem of welfare economics that isolates conditions under which a competitive equilibrium allocation is Pareto optimal.[5] The condition that is violated by equilibrium L is the assumption that the value of the aggregate endowment at the equilibrium prices is finite.[6]

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[4] See Karl Shell (1971) for an investigation that characterizes why some competitive equilibria in overlapping generations models fail to be Pareto optimal. As maintained by Shell, "the double infinity of traders and dated commodities allows for competitive equilibria that are not Pareto-optimal [and] the fact that generations do not meet is not essential."

[5] See Mas-Colell, Whinston, and Green (1995) and Debreu (1954).

[6] Note that if the horizon of the economy were finite, then the counterpart of equilibrium H would not exist and the allocation of the counterpart of equilibrium L would be Pareto optimal.

10.3.4. Nonstationary equilibria

Our example economy has more equilibria. To construct them, we use an offer curve to summarize preferences and optimal consumption decisions. We describe a graphical apparatus proposed by David Gale (1973) and used to great advantage by William Brock (1990).

DEFINITION: Agent i's *offer curve* is a locus of (c_i^i, c_{i+1}^i) that solves

$$\max_{\{c_i^i, c_{i+1}^i\}} U^i(\mathbf{c}^i)$$

subject to

$$c_i^i + \alpha_i c_{i+1}^i \leq y_i^i + \alpha_i y_{i+1}^i.$$

Here $\alpha_i \equiv \frac{q_{i+1}^0}{q_i^0}$, the reciprocal of the one-period gross rate of return from period i to $i+1$, is a parameter that we shall use to sweep out a (c_i^i, c_{i+1}^i) locus.

Evidently, points (c_i^i, c_{i+1}^i) on the offer curve simultaneously solve the following pair of equations:

$$c_i^i + \alpha_i c_{i+1}^i = y_i^i + \alpha_i y_{i+1}^i \tag{10.3.7a}$$

$$\frac{u'(c_{i+1}^i)}{u'(c_i^i)} = \alpha_i \tag{10.3.7b}$$

for different values of the relative price $\alpha_i > 0$. We denote the offer curve swept out by $\alpha_i > 0$ by

$$\psi(c_i^i, c_{i+1}^i) = 0.$$

The offer curve is illustrated in Figure 10.3.1. We trace it out by varying α_i in the consumer's problem and reading off tangency points between the consumer's indifference curve and the budget line. So at each point on the offer curve, there is an indifference curve with a tangent at that point which is a straight line that can be extended to go through the consumer's endowment point. Hence, the locus of the offer curve depends on the endowment vector and lies above the indifference curve through the endowment point. Since the consumer can always choose to consume his endowment, any optimally chosen consumption (c_i^i, c_{i+1}^i) could never lie on an indifference curve lower than the indifference curve through the endowment point.

By relabeling axes, in the same graph we can plot a consumption possiblity frontier

$$c_i^i + c_i^{i-1} \leq y_i^i + y_i^{i-1}$$

that bounds feasible allocations at time i, which we call the *feasibility line.* Note that while the offer curve restricts a life-time consumption profile (c_i^i, c_{i+1}^i) for one person born at time i, the feasibility line involves pairs (c_i^i, c_i^{i-1}) of consumptions assigned to a young and old person at time i.[7]

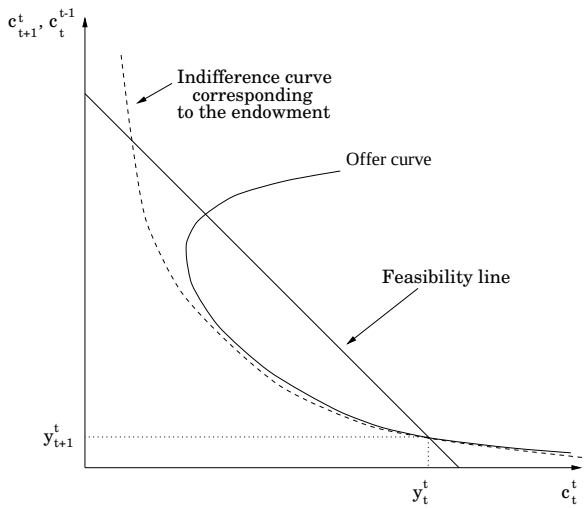

Figure 10.3.1: The offer curve and feasibility line.

Following Gale (1973), for the important special case in which $(y_i^i, y_i^{i-1}) = (y_y, y_o)$ for all $i \geq 0$, we can use the offer curve and a straight line depicting feasibility in the (c_i^i, c_i^{i-1}) plane to compute equilibrium allocations and prices. The assumption of stationary endowments means that the feasibility line is the same in all periods $i \geq 0$. As explained earlier, the initial old agent consumes his endowment in an equilibrium, $c_0^{-1} = y_o$. Regarding the remaining equilibrium allocation, there is a multiplicity of possible outcomes since

[7] Given our assumptions on preferences and endowments, the conscientious reader will note that Figure 10.3.1 appears distorted because the offer curve really ought to intersect the feasibility line along the 45 degree line with $c_t^t = c_{t+1}^t$, i.e., at the allocation affiliated with equilibrium H above.

$y_o < y_y$. Specifically, we can initiate an equilibrium computation by choosing $c_0^0 \in ((y_y + y_o)/2, y_y)$.[8] As we shall see, for each different c_0^0, a different competitive equilibrium prevails. Thus, we can find an equilibrium allocation by solving a pair of nonlinear difference equations. For $i \geq 0$, the equations are

$$\psi(c_i^i, c_{i+1}^i) = 0, \tag{10.3.8a}$$

$$c_{i+1}^{i+1} + c_{i+1}^i = y_y + y_o. \tag{10.3.8b}$$

Equation (10.3.8a) states that the allocation to an agent born at time i is on the offer curve. Equation (10.3.8b) asserts that the allocation at time $i+1$ is feasible.[9] So we take c_0^0 as an initial condition and then compute an equilibrium allocation via the following steps:

(**a**) Solve agent 0's offer curve (10.3.8a) for c_1^0.

(**b**) Solve feasibility condition (10.3.8b) for c_1^1.

(**c**) Iterate on these steps indefinitely and hope for convergence.

Dropped "optional step (d)" on gift giving to the initial old.

After an allocation has been computed, a time 0 trading equilibrium price system can be computed from

$$q_{i+1}^0 = q_i^0 \frac{u'(c_{i+1}^i)}{u'(c_i^i)}$$

for all $i \geq 0$, given the numeraire $q_0^0 = 1$.

[8] When setting c_0^0 equal to one of the two boundary points, $(y_y + y_o)/2$ or y_y, the resulting equilibria would be the counterparts to the two stationary equilibria in section 10.3.2.

[9] Feasibility in period $i = 0$ is assured by our choice of $c_0^0 < y_y$ and the initial old agent consuming his endowment in any equilibrium, $c_0^{-1} = y_o$, i.e., equation (10.3.1b) is satisfied with $\Omega_0 > 0$.

10.3.5. Computing equilibria

We now describe David Gale's (1973) equilibrium computation machine that consists of a graph like that illustrated in Figure 10.3.2. Notice that y_t^t and c_t^t are on the ordinate axis while y_{t+1}^t and both c_{t+1}^t and c_t^{t-1} are on the coordinate axis. The straight feasibility line with slope -1 goes through endowment point (y_t^t, y_{t+1}^t) and plots feasible pairs (c_t^t, c_t^{t-1}) that can be allocated to people alive at time t. The curved offer curve also goes through the endowment point (y_t^t, y_{t+1}^t) and plots pairs (c_t^t, c_{t+1}^t) that a person born at t optimally chooses as a function of the relative price α_t implicitly sweeping out the offer curve.

Applying the algorithm in the preceding section, we start with a proposed c_0^0, a time 0 allocation to the initial young. The candidate time 0 equilibrium allocation is feasible in accordance with footnote 9, but the time 0 young will choose c_0^0 only if the price α_0 is such that (c_0^0, c_1^0) lies on the offer curve. Therefore, we choose c_1^0 from the point on the offer curve that cuts a vertical line through c_0^0. Then we proceed to find c_1^1 from the intersection of a horizontal line through c_1^0 and the feasibility line. We continue recursively in this way, choosing c_i^i at the intersection of the feasibility line with a horizontal line through c_i^{i-1}, then choosing c_{i+1}^i at the intersection of a vertical line through c_i^i and the offer curve. We can construct a sequence of α_i's from the slope of a straight line through the endowment point and the sequence of (c_i^i, c_{i+1}^i) pairs that lie on the offer curve.

If the offer curve has the shape drawn in Figure 10.3.2, any c_0^0 between the two intersections of the offer curve and the feasibility line is an equilibrium setting of c_0^0. Each such c_0^0 is associated with a distinct allocation $\{c_i^i, c_{i+1}^i\}_{i\geq 0}$ and price system $\{\alpha_i\}_{i\geq 0}$, while consumption of the initial old agent is the same across all equilibria, $c_0^{-1} = y_o$. If c_0^0 is chosen to be the lower boundary point for that set, the economy will simply remain in that *high*-interest-rate stationary equilibrium. For all the other choices of c_0^0, the economy will converge to the *low*-interest-rate stationary equilibrium allocation and interest rate.

Log utility: Suppose that $u(c) = \ln c$ and that the endowment is described by equations (10.3.5) with $\epsilon \in (0, .5)$. After solving equations (10.3.7) for that specification, the offer curve is given by the recursions

$$c_i^i = .5(1 - \epsilon + \alpha_i \epsilon), \tag{10.3.9a}$$

$$c_{i+1}^i = \frac{1}{\alpha_i} c_i^i. \tag{10.3.9b}$$

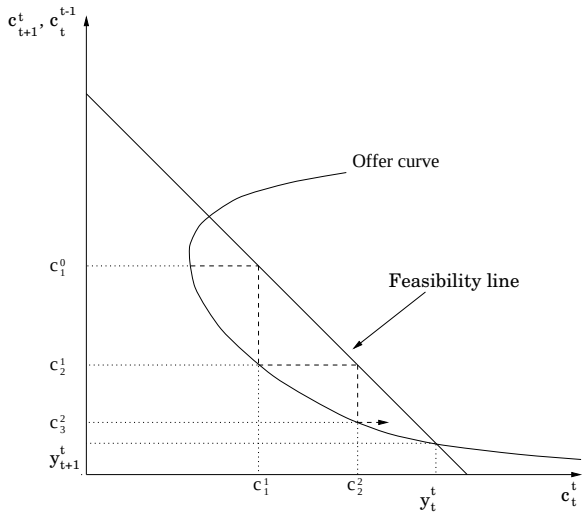

Figure 10.3.2: A nonstationary equilibrium allocation.

Hence, feasibility at $i \geq 1$ and the offer curves then imply

$$\frac{1}{2}(1-\epsilon+\alpha_i\epsilon)+\frac{1}{2\alpha_{i-1}}(1-\epsilon+\alpha_{i-1}\epsilon)=1. \tag{10.3.10}$$

This implies the difference equation

$$\alpha_i=\epsilon^{-1}-\frac{\epsilon^{-1}-1}{\alpha_{i-1}}. \tag{10.3.11}$$

To find the stationary equilibria of this economy, we let $\alpha_i=\alpha$ for all i in difference equation (10.3.11) and after rearranging,

$$\alpha^2-\epsilon^{-1}\alpha+\left(\epsilon^{-1}-1\right)=0. \tag{10.3.12}$$

Using the quadratic formula, the equilibrium prices in two stationary equilibria are:

$$\alpha=\frac{\epsilon^{-1}\pm\sqrt{\epsilon^{-2}-4\left(\epsilon^{-1}-1\right)}}{2}=\frac{\epsilon^{-1}\pm\sqrt{\left(\epsilon^{-1}-2\right)^2}}{2}=\begin{cases}1,\\ \dfrac{1-\epsilon}{\epsilon}\end{cases} \tag{10.3.13}$$

The two equilibria are the log-utility versions of the two stationary equilibria in section 10.3.2, $\alpha_H=1$ and $\alpha_L=u'(\epsilon)/u'(1-\epsilon)=(1-\epsilon)/\epsilon>1$. To initiate the

Removed "alpha<1 implies that the tail is an unbounded negative sequence". Actually, the difference equation turns negative once before jumping to a large positive value which then converges to the low-interest-rate stationary equilibrium (!) but from the wrong side. We probably don't want to talk about these weird non-equilibrium dynamics.

computation of a nonstationary equilibrium using difference equation (10.3.12), we choose $\alpha_0 \in (\alpha_H, \alpha_L)$, which is tantamount to choosing $c_0^0 \in (.5, 1-\epsilon)$ when using the algorithm in section 10.3.4. The limiting value of $\{\alpha_i\}$ for any nonstationary equilibrium is α_L. Hence, the autarkic equilibrium α_L is stable, while the stationary equilibrium $\alpha_H = 1$ is not.

10.4. Sequential trading

We now alter the trading arrangement to bring it into line with more common presentations of the overlapping generations model. We abandon the time 0, complete markets trading arrangement and replace it with sequential trading. In our model with two-period lived agents, the only equilibrium is then autarky since there is no double coincidence of wants. While a young agent might want to buy or sell goods in exchange for goods next period, a current old agent will not be around next period. Hence, the unique equilibrium price system is the one that eradicates all incentives to trade, $\alpha_t = u'(y_{t+1}^t)/u'(y_t^t)$ for all $t \geq 0$.

To create opportunities for trade across generations, we introduce a government that at time 0 issues a zero-dividend infinitely durable asset in the form of fiat money or unbacked government debt. Under the auxiliary assumption that the government gives this asset to the initial old agent, we will show that all the equilibria of the time 0 trading economy can reemerge as sequential trading equilibria but for one difference. Any goods that the clearinghouse had earlier left to perish at time 0, $\Omega_0 \geq 0$, will no longer be wasted but be consumed by the initial old agent. We study sequential trading equilibria under the common notion of a Walrasian auctioneer who announces equilibrium prices at which the goods and asset markets clear in every period.

The assumption that the government-issued asset is given to the initial old generation is critical for the described equivalence of outcomes under time 0 trading and sequential trading. Any giveaway program to the initial old generation has no effects on the equilibrium price system, since those old agents optimally consume all their wealth in period 0. If the assets had instead been given to the initial young generation, it would have changed their offer curves and hence, affected equilibrium prices.